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MIL-T-29185B
19 September 1989
SUPERSEDING
MIL-T-29185A(YD)
9 May 1984

MILITARY SPECIFICATION

TRUCK, FIREFIGHTING: BRUSH/GRASS, 18,000 POUNDS GVW, 4 BY 4, COMMERCIAL CHASSIS

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers a gasoline or diesel engine-driven, 4-wheel, 4-wheel-drive, 250 gallons per minute (gpm) pumping capacity, fire-fighting truck having pump and roll capability to fight brush and grass fires.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

W-B-131 - Battery, Storage, Vehicular, Ignition, Lighting and Starting.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4210

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MILITARY

MIL-I-45208 - Inspection System Requirements.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.
FED-STD-297 - Rustproofing of Commercial (Nontactical) Vehicles.
FED-STD-595 - Colors (Requirements for Individual Color Chips).

MILITARY

MIL-STD-281 - Automobiles, Trucks, Truck-Tractors, and Trailer Dollies; Preservation and Packaging of.
MIL-STD-1223 - Nontactical Wheeled Vehicles Treatment, Painting, Identification Marking, and Data Plate Standards.

(Unless otherwise indicated, copies of federal and military specifications and standards are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

(Copies of specifications, standards and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.1.2 Other Government documents. The following other Government documents forms a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

ENVIRONMENTAL PROTECTION AGENCY (EPA)

Title 45, CFR, Part 1201 Subpart A - Motor Vehicle Air Pollution Standards.
Interstate Motor Carrier Noise Emission Standards.

(Application for copies should be addressed to the Public Affairs Office, Environmental Protection Agency, Rockville, MD 20852; or CFR, Title 40 should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

DEPARTMENT OF TRANSPORTATION (DoT)

Federal Motor Vehicle Safety Standards and Regulations.
Federal Motor Carrier Safety Regulations.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

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2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are Department of Defense (DoD) adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents which is current on the date of the solicitation (see 6.2).

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 1901 - Automobile Fire Apparatus.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE J534 - Lubrication Fittings.
SAE J537 - Storage Batteries.
SAE J551 - Limits and Methods of Measurements of Radio Interference Characteristics of Vehicle and Devices (20-1000 MHZ).
SAE J1349 - Engine Power Test Code - Spark Ignition and Diesel.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

TIRE AND RIM ASSOCIATION, INC. (TRA)

TRA Yearbook.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Akron, Suite 304, OH 44313.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification takes precedence. Nothing in this specification, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The truck shall be a 4-wheel, 4-wheel drive, gasoline or diesel engine-driven truck cab-and-chassis having a gross vehicle weight (gvw) rating of not less than 18,000 pounds (lb). The truck shall be fitted with a fire truck body, a 250 gpm minimum booster type midship mounted fire pump

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driven by a power takeoff (pto) mounted on the transmission to provide pump and roll capability, a 500 gallon (gal) removable water tank, hose beds, booster hose reel, operator's control panel for fire pump, and such other equipment and accessories as specified herein.

3.2 First article. When specified (see 6.2), the contractor shall furnish a truck for first article inspection and approval (see 4.2.1 and 6.4).

* 3.2.1 Inspection system requirements. When specified (see 6.2), quality assurance inspection and test necessary to substantiate product conformance to contract specifications shall be in accordance with MIL-I-45208.

3.3 Standard truck, components, and accessories. Except as otherwise specified in 3.3.1 through 3.3.9, the truck, components, and accessories shall be standard or optional items, which meet or exceed the requirements of this specification.

3.3.1 Drainplugs. All drainplugs installed throughout the truck for the purpose of draining lubricants shall be of the permanent magnet type.

3.3.2 Heavy-duty cooling system. A heavy-duty cooling system shall be provided. The heavy-duty cooling system shall be of the pressurized type of sufficient capacity to maintain a cooling liquid temperature range of -20 degrees Fahrenheit (°F) to +125°F during operating conditions of either stationary pumping or mobile performance as specified.

3.3.3 Coolant system and indicators. The coolant system shall include a deaeration system, a surge tank, or a coolant recovery reservoir of not less than 2-quart capacity. A high coolant temperature or low coolant level alarm buzzer and red indicator warning light shall be provided on the cab dash or instrument panel.

3.3.3.1 Heat exchanger. An auxiliary heat exchanger shall be provided, which shall permit delivery of water from the fire pump to the heat exchanger for cooling of the engine liquid coolant without admixture. The heat exchanger shall not be connected to the cab heater circuit. The heat exchanger shall be the type and size approved by the pump manufacturer and may be separate from the radiator. The discharge water from the heat exchanger shall be piped into the suction side of the fire pump. The size of the heat exchange discharge line shall be large enough to eliminate back pressure on the heat exchanger. The flow of water from the fire pump to the heat exchanger shall be controlled by a valve, with the valve control extended to the fire pump operator's control panel.

3.3.3.2 Drain valves. Water drain valves shall be installed at the low point of the cooling and heat exchanger systems and at any other points necessary to completely drain the systems. The drain valves shall be of a type that will remain closed under the most severe operating conditions, to be encountered by the fire truck. Drain lines shall be furnished and shall discharge to the ground at a point not above the bottom of the chassis frame.

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3.3.4 Front bumper and towing devices. A heavy duty, full width, front bumper and two tow hooks shall be provided at the front of the truck. The two front tow hooks shall be mounted to the frame or rigid members which are attached to the frame. Two recessed, 3-inch inside diameter (id), towing rings or comparable U-bolts, shall be provided at the rear of the truck, and mounted directly to the chassis side rails.

3.3.5 Brush guard. A brush guard shall be provided. The brush guard shall be as wide as the widest part of the front fenders, and shall be as high as the hood. The brush guard shall be at least 4-point suspension mounted and shall be removable.

3.3.6 Engine hour meter. An engine hour meter of the oil pressure or electrical type, having a totalizing mechanism of not less than 9,999 hours, shall be furnished for the truck chassis engine to register accurately the number of hours of operating time. The meter shall be of rugged construction to insure continuous trouble-free performance under severe operating conditions. Engine hour meter shall be mounted on the cab instrument panel or in the engine compartment in a readable location.

3.3.7 Radio interference suppression. The vehicle shall be suppressed to limit the electromagnetic radiation in accordance with SAE J551. Any body equipment emitting radiation shall be suppressed to the same level as the vehicle chassis.

3.3.8 Air pollution control. Trucks shall comply with the EPA regulations governing control of air pollution from new motor vehicles and new motor vehicle engines in effect on the date of manufacture.

3.3.9 Sound level. The exterior sound level produced by the truck shall not exceed the noise limits established by the EPA. The interior sound level shall be in accordance with DoT Federal Motor Carrier Safety Regulations, section 393.94.

3.4 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.5 Performance. The fully equipped truck loaded as specified in 3.6.2, shall be capable of performing the following:

a. Parking brake holding the truck on a 30 percent grade, in both ascending and descending positions, with the fire pump operating and discharging a full tank of water at a flow rate of not less than 96 gpm at a pressure of not less than 100 pounds-force per square inch (psi) through the 1-1/2-inch hose.
b. Service brakes holding the truck on a 30 percent grade in both ascending and descending positions.
c. Service brakes bringing the truck to a stop within a braking distance of 30 feet (ft) from a speed of 20 miles per hour (mph) on a dry, smooth, level surface free from loose material.

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d. Maintaining a maximum speed of not less than 55 mph on dry, level, paved roadway.
e. Accelerating on a level road from a standing start to a speed of 35 mph in not more than 20 seconds.
f. Maintaining a sustained speed when ascending a 30 percent grade in low speed range.
g. There shall be no evidence of body distortion, water leakage, tire and body contact, malfunction of components, irregular chassis noise, vibration or sway when subjected to a road test of 50 miles over paved roadway at not less than 55 mph, 25 miles over graded dirt roadway at not less than 25 mph, and 25 miles off highway over cross-country terrain at speeds up to 15 mph with ground and grade conditions requiring 4-wheel drive.
h. In low speed range, provide uninterrupted discharging of full water tank of not less than 120 gpm at 100 psi nozzle pressure through two 1-1/2-inch hose lines during continuous forward motion over cross-country terrain at a truck speed of not less than 3 mph.
i. Negotiating side slopes up to 20 percent on a surface reasonably hard and free from loose material.

3.6 General design.

3.6.1 Federal motor vehicle safety standards. The truck shall comply with all DoT Federal Motor Vehicle Safety Standards in effect on the date of manufacture.

3.6.2 Weight rating. The manufacturer's gvw for the truck chassis furnished shall not be less than 18,000 lbs. The total gross weight of the complete truck shall not exceed the manufacturer's gvw rating when fully equipped as specified herein, including all specified firefighting equipment, serviced with fuel, water, and lubricants; loaded with 500 gal of water in the water tank; not less than a 2,000 lb simulated load in the hose beds, and a crew of three men in the cab weighting not less than 175 lbs each or equivalent dead weight. Weight distribution ratio between the front and rear axle shall be such as to insure positive traction at all four wheels and positive steering at the front wheels under all operating conditions specified herein. Percentage of gvw at front axle shall be not less than 25 percent or more than 30 percent. The application of weights to the chassis for the purpose of meeting the gvw distribution requirements between front and rear axles is not acceptable.

3.6.3 Cab-to-axle. The cab-to-axle dimension shall be not less than 84 inches.

3.6.4 Dimensions. The truck overall dimensions shall be not more than the following:

a. Width - 96 inches.
b. Length - 276 inches.
c. Height - 108 inches.

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3.7 Chassis components. The truck-chassis shall be a two axle wheel, 4-wheel drive, front wheel steer, dual rear wheel, with conventional type cab.

3.7.1 Engine. The engine furnished shall be the chassis manufacturer's standard, or optional engine for the commercial model truck which meets or exceeds the requirements of this document.

3.7.1.1 Gasoline engine. When specified (see 3.7.1.2), the truck shall be equipped with a liquid cooled, internal combustion, four-stroke cycle gasoline engine with not less than eight cylinders. The engine furnished shall produce the required truck performance using unleaded gasoline with a research octane rating of 91. At a speed not more than the manufacturer's recommended operating speed. The engine shall be capable of warranted operation on unleaded fuel when operated in accordance with the operator's manual. Maximum net engine torque shall be not less than 320 pound-feet. Engine net torque ratings and net horsepower (hp) shall be determined in accordance with SAE J1349.

3.7.1.2 Diesel engine. Unless otherwise specified (see 6.2), the truck shall be equipped with a liquid cooled, compression ignition, two-stroke or four-stroke cycle diesel engine. Engine net hp figures used in performance prediction calculations shall be determined in accordance with SAE J1349. A fan clutch to reduce fan speed automatically, when not required for engine cooling, shall be provided.

3.7.1.3 Lubrication system. The engine lubrication system shall be the full forced feed type. A low oil pressure warning device shall be provided and located on the fire pump operator's control panel. A full flow type oil filter shall be furnished.

3.7.1.4 Governor. Manufacturer's current standard type engine governor shall be furnished and set and sealed to limit engine to maximum recommended operating speed.

3.7.2 Electrical equipment. The truck shall be provided with a complete 12-volt (V) negative ground starting and lighting system. Electrical equipment, including all truck lights and wiring shall conform to DoT Federal Motor Carrier Safety Regulations, sections 393.12, 393.19, 393.20, 393.22, 393.24, 393.25, 393.27 through 393.31, and 393.33, except as specified herein. Reflectors shall conform to DoT Federal Motor Carrier Safety Regulations, section 393.26. Wiring shall be weatherproof, resistant to moisture and fungus, and color-coded or numbered for ready identification. Wiring identification shall be consistent throughout the truck. Two spare, single pole, 15-ampere (A) breakers shall be furnished for future radio use. The truck chassis electrical circuits and the truck body electrical circuits shall be separate circuits. The truck body electrical circuits, excluding truck chassis electrical circuits and the truck body electrical circuits shall be separate circuits. The truck body electrical circuits, excluding truck chassis electrical circuits, shall be provided with overload protective devices of the automatic reset, overload circuit-breaker type.

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3.7.2.1 Ignition system. Ignition system of 12V potential shall be furnished. Alternator of not less than 145 A rated capacity, and which provides not less than 40 A direct current output at normal engine idle speed shall be furnished.

3.7.2.2 Battery. The battery shall be 12V potential with negative ground. The total reserve capacity rating shall be not less than 100 minutes and the total cold cranking rating shall be not less than 450 A at 0°F, both measured in accordance with SAE J537. The battery shall be of the maintenance free type with the maintenance free characteristics listed in W-B-131.

3.7.2.3 Lighting. All truck lights, reflectors, and wiring shall be as specified herein and shall conform to DoT Federal Motor Carrier Safety Regulations, sections 393.12, 393.19 through 393.31, and 393.33. Lights and reflectors shall not be mounted on vertical surfaces of rub rails or bumpers.

3.7.2.3.1 Lights. All lights shall be mounted in such a manner as to prevent being obscured by riding personnel or mounted equipment. All metal rims, shells, mounting arms, and bases of all exterior lights, except headlights rims, shall be chromium plated or polished stainless steel. Stoplights and taillights shall be flush mounted at the rear of the body. All warning lights shall operate off a single switch, mounted on the cab instrument panel and shall be provided with a red pilot light and marked to indicate function. The following additional lighting shall be furnished and shall be provided with overload protection:

a. Two combination spotlight and red flasher lights, Unity Model 10300 or equal. The lights shall be installed on the windshield posts with controls extended inside the cab for ON-OFF control and movement of the lamp housing through a horizontal included angle of at least 270°, and a vertical angle of at least 90° above and 35° below horizontal.
b. Two adjustable, sealed beam, hose pick-up lights with weatherproof switches mounted on rear handrail supports.
c. Two automatic backup lights, not less than 6 inches in diameter, flush mounted at rear of body.
d. Pump control panel lights.
e. Two 6-inch sealed beam, red alternating flashing warning lights shall be mounted on the rear handrail supports adjacent to the hose pick-up lights. The lamps shall have a minimum output rating of 200 candlepower. The lamps shall have a combined minimum flash rate of 80 flashes per minute.
f. Two low mounted red flashing lights shall be mounted on the front of the truck. The warning lights shall be installed adjacent to or over the headlights. The warning lights shall be sealed beam lamps, 6 inches in diameter, with a minimum output rating of 200 candle-power. The lights shall have a combined minimum flash rate of 80 flashes per minute.

3.7.2.4 Warning light. Manufacturer's standard cab roof mounted lightbar, with built-in public address (PA) system speaker, shall be furnished. The sealed beam lamps shall be high intensity, red and white in color. The light system housing shall be clear. The speaker shall be as specified in 3.7.2.4.1.

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3.7.2.4.1 Electronic siren, public address, and radio amplifier system. An electronic siren system with selector controls for the siren, PA, and radio amplification through an external speaker shall be provided. System sound intensity shall be not less than 98 decibels at a distance of 100 ft from the speaker. The speaker shall be at least 100 watt (W). The amplifier and controls shall be installed inside the vehicle in a location accessible to the driver and assistant driver. A microphone of the magnetic noise canceling type, connected by a coiled cord, shall be provided and located in a position convenient for use. Microphone stowage shall be provided on the instrument panel. The system shall permit broadcasting routine and emergency announcements from an onboard Government installed radio. The amplifier shall include volume control and selection of "Radio," "PA," "Manual," "Yelp," "Wail," and "HiLo" (European) modes. Switch positions shall be identified. A 2-way radio audio tap cable for rebroadcasting 2-way radio messages through the PA system shall be furnished. The speaker shall be fitted with a weather-proof housing and shall be suitable for installation in the lightbar specified in 3.7.2.4.

3.7.3 Fuel system. Fuel system shall conform to DoT Federal Motor Carrier Safety Regulations, sections 393.65 and 393.67.

3.7.3.1 Air cleaner. Manufacturer's standard dry type air cleaner shall be furnished.

3.7.3.2 Fuel tank(s). The fuel tank(s) shall be of not less than 40 gal total capacity. Tank and fill piping shall be so placed as to be protected from mechanical injury.

3.7.3.3 Exhaust system. Exhaust system shall conform to DoT Federal Motor Carrier Safety Regulations, section 393.83. Where necessary, exhaust pipe shall be insulated and shielded. A spark arresting muffler or a spark arrester shall be provided. Muffler and tailpipe shall be aluminized steel. The exhaust system discharge shall be directed away from the fire pump operator's control panel area.

3.7.3.4 Fuel pump. The fuel system shall be provided with either a mechanical or electric fuel pump.

3.7.3.5 Fuel filter. The fuel system shall be provided with a fuel filter installed at the tank and in the fuel line.

3.7.4 Automatic transmission. The truck shall be provided with an automatic transmission of a fixed ratio type and shall provide continuous drive. The fixed ratio shall include a hydraulic torque converter and not less than four forward speeds. The normal driving range selector position shall provide not less than four gear ratios without movement of the selector. A drive gear for the pto shall be provided. Input torque capacity of the transmission shall be at least equal to the maximum torque delivered by the engine.

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3.7.4.1 Power takeoff. A transmission drive pto shall be provided to drive the fire pump. The pto and pump gear ratio shall be selected to provide the required performance in pump and roll operations. The pto controls to place the pump in gear shall be located in the chassis cab. A caution plate reading: "DO NOT OPERATE TRUCK AT HIGHWAY SPEEDS WITH POWER TAKEOFF ENGAGED" Shall be installed on the instrument panel.

3.7.5 Transfer case. A 2-speed type transfer case shall be provided. The transfer case shall provide for driver selection of either 2-wheel or 4-wheel drive.

3.7.6 Drive line component. Drive line components shall be adequate to transmit the maximum delivered torque of the engine, as developed through the maximum gear train reduction.

* 3.7.7 Drive hubs. Manufacturers standard drive hubs, for disengaging front wheels from power train shall be provided.

3.7.8 Frame. The chassis frame shall be the manufacturer's standard welded, riveted, or bolted construction, and shall be rated for operation at rated gvw. The frame shall be reinforced. Reinforcements shall extend at least from the rear of the front-suspension, rear hanger bracket, to the front of the rear-spring, front hanger bracket. When operating the fully equipped and loaded truck under conditions specified herein, the frame shall show no evidence of permanent deformation or distortion.

3.7.9 Suspension. The truck shall be equipped with suspension system with components having a rated capacity at least equal to the load imposed on each member, plus at least 10 percent, measured at the ground, with truck loaded to specified gvw. When suspension is rated at the spring pads, unsprung weight shall be deducted. The truck shall be equipped with hydraulic double-acting shock absorbers on the front axle. The chassis manufacturer's standard auxiliary rear springs shall be furnished.

3.7.10 Axles. The front axle shall have a rated capacity of not less than 5,000 lbs., and the rear axle shall have a rated capacity of not less than 15,000 lbs. Rear axle shall be provided with positive-traction or limited-slip differential.

3.7.11 Wheels, rims, tires, and tubes. The truck shall have single front and dual rear wheels. Rim and tire ratings shall conform to TRA recommendations. Tires shall be tube or tubeless type of not less than 100 level quality with standard commercial mud and snow tread. Steel belted radial tires shall be furnished. Tire flaps shall be provided for tube type tires in accordance with REA recommendations. All tires furnished on the truck shall be of the same size. The rated individual tire capacity at 55 mph highway conditions shall be not less than the maximum wheel loading imposed with truck loaded to rated gvw. Wheel and tire assemblies shall be dynamically balanced at simulated speeds up to 55 mph. A spare wheel with mounted tire shall be furnished and stowed in the hose bed for shipment. Disk type wheels shall be provided.

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3.7.12 Brakes. Brakes shall conform to DoT Federal Motor Carrier Safety Regulations, sections 393.40 through 393.42, and 393.45 through 393.51, and as specified herein.

3.7.12.1 Service brakes. The truck shall be equipped with a power assisted, hydraulic actuated, four wheel service brakes, or complete air brake system is acceptable.

3.7.12.2 Parking brakes. The truck shall be equipped with a parking brake.

3.7.13 Steering. The truck shall be equipped with power assisted steering. Mechanical type steering capability shall be available in event of loss of power assist steering.

3.7.14 Cab. The cab shall be all metal construction and shall provide unobstructed seating for three men. The engine and driver's compartment shall be separated by an insulated firewall. Cab shall have upholstered, full-width, adjustable seat. Cab shall be provided with booths for all levers and controls extending to the exterior of the cab. The cab doors shall be equipped with locks, operable from inside the cab through mechanical linkage, and with at least the curb side door equipped with external, key operated lock. Safety grips shall be provided on each side of the cab to assist personnel in climbing into the truck cab. Fastenings and two pairs of seat belts, and inside rear-view mirror shall be furnished. Safety glass shall be provided throughout the cab. Dual sun visors shall be furnished. A dome light shall be provided.

3.7.15 Windshield wipers and washers. The truck shall be equipped with dual windshield wipers and windshield washers. Windshield wipers shall be multispeed type and operated by electric motor(s). Windshield washers equipped with a plastic container of at least 2 quart capacity for cleaning fluid shall be furnished.

3.7.16 Heater and defroster. A hot water heater shall be provided. The heater shall have fresh air intakes. Discharge outlets shall be provided to direct heated air to floor and to defroster louvers. The heater shall be complete with blower and directional controls convenient to the driver.

3.7.17 Controls and operating mechanism. All controls and operating mechanisms shall be located for left-hand drive. Controls shall be complete and conveniently operable by the driver. Level controls shall be designed and located to permit easy entrance and exit of the operator to and from the driver's compartment. Instruments and controls shall be identified as to their function and installed in a manner to facilitate removal and servicing. Instruments shall be dash mounted.

3.7.18 Back-up warning device. The fire truck shall be provided with an automatically activated, audible, pulsating, signaling warning device of at least 100 decibels at 10 ft, to caution personnel when fire truck is in reverse gear operation.

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3.7.19 Accessories and equipment. Chassis equipment shall be complete with all accessories furnished as standard equipment by the manufacturer. The following minimum equipment shall be furnished:

a. Outside rear view mirrors. The truck shall have an exterior mounted rear view mirror on both the right and left sides of the cab. Each mirror shall have at least three supporting arms and heavy duty spring loaded rachet type mounting brackets. Combination flat and convex mirrors enclosed in a common housing shall be furnished. Each combination mirror shall have not less than 50 square inches of flat reflective area and convex surface having not less than 20 square inches of reflective area.
b. Pump pressure gage (0-400 psi).
c. Ammeter gage or charging indicator.
d. Fuel gage.
e. Oil pressure gage.
f. Engine temperature gage.
g. Speedometer with recording odometer.
h. Ash receptacle.
i. Dual sun visors.
j. Door mounted armrest on driver's side.
k. Driver's compartment ventilator other than window.
l. Manufacturer's standard electric horn.
m. Electric tachometer. Range 0-4,000 revolutions per minute (rpm).

3.8 Body. The truck shall be equipped with a body which shall incorporate a fire pump housing and stowage bin, fire pump operator's control panel, removable water tank, hose bed, equipment cabinets, rear platform, booster hose reel and firefighting equipment. The body shall be fabricated from aluminum or cold roll steel and shall be of welded construction. The body shall be reinforced at all points where hose and ladder clamps and hangers are attached. The body shall be free from rough and sharp edges.

3.9 Fire pump. The fire pump shall be a booster type midship mounted centrifugal pump that is pto driven from the transmission. The pump drive shall be controllable from the cab. The pump furnished shall conform to the requirements of NFPA 1901. Controls for stationary vehicle pumping operations and "pump-and-roll" operations shall be operable from the driver's position. Controls for pump engage and disengage operation shall be of the positive control type. The pump impeller shaft shall be of corrosion-resistant steel. The fire pump shall have the manufacturer's standard high nickel content cast iron casing. The pump drive, gearing, and housing shall be such that at rated capacity and pressure, the temperature rise in the gear oil will not exceed the maximum temperature recommended by the pump manufacturer for continuous operation. When required, oil cooling shall be provided by a water jacket or cooling coils in the gearcase constructed so as not to be damaged by freezing. All shafts in the pump gear train shall turn in antifriction bearings. All pump-shafts glands or mechanical seals shall be adequate for service under operating conditions specified herein. The pump shall withstand a hydrostatic pressure of 350 psi for a period of 10 minutes. The pump shall withstand a discharge pressure of not less than 400 psi. Drain shall withstand a discharge pressure of ot less than 400 psi. Drain cocks,

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push-pull or rotary turn type, accessible at the pump operator's position and arranged so that the open and closed position is clearly indicated, shall be provided to drain the pump and all water-carrying lines. Drain cocks shall be of the water service type and shall be equipped with positive stops. Pump drain lines shall be furnished and shall discharge to the ground at a point not above the bottom of the chassis frame. The fire pump shall be provided with piping and valve control at the pump operator's panel for filling the water tank while drafting water. All threaded connections shall be National pipe (thread) and shall conform to FED-STD-H28. The pump, when drafting at a minimum lift of 10 feet through 20 feet of 3-inch suction hose shall:

a. Deliver a minimum of 250 gpm at 150 psi discharge pressure at less than the manufacturer's recommended governed speed of the engine.
b. Deliver a minimum of 100 gpm at 250 psi discharge pressure at less than the manufacturer's recommended governed speed of the engine.
c. Deliver each flow rate specified above at an additional 10 percent discharge gage pressure.
d. The pump, when dry, shall take suction and discharge water within 30 seconds, with a lift of 10 feet through 20 feet of 3-inch suction hose.
e. The pumping requirements specified herein shall be accomplished at all altitudes up to 2,000 feet above sea level and without evidence of overheating, vaporlock, excessive vibration, or any other malfunction.
f. The fire pump shall also deliver not less than 250 gpm from the water tank and shall maintain the rate of flow for not less than 80 percent of the rated capacity of the tank.
g. The fire pump shall also deliver not less than 60 gpm at a minimum 100 psi from the water tank through the booster hose nozzle during stationary and pump and roll operations. Also, during pump and roll operations, the fire pump shall deliver the flow rates and requirements as specified in 3.5.i.

3.9.1 Priming pump. The priming pump shall be of the rotary type, 12 V, motor driven, complete with lubricating oil reservoir and necessary valves, operated from the pump control panel. The priming pump shall develop a vacuum of 22 inches of mercury (Hg) at an altitude up to 1,000 feet above sea level in not more than 30 seconds.

3.9.2 Suction connections. The fire truck shall be provided with one gated 3-inch suction connection located on the fire pump operator's control panel. The gated valve shall be the pump manufacturer's standard corrosion-resistant, balanced ball type for quarter-turn operation. The valve shall be operated and locked by means of an individual control handle as a single-handed cap. The suction connection shall be equipped with a removable and accessible strainer; shall have a machined end with mirror finish to permit hand tight connections; shall provide a seat for suction hose gaskets and shall be threaded with American National fire hose coupling thread (NH). All piping from the suction connection shall be 3-inch id.

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3.9.3 Discharge connections. The fire truck shall be provided with one 2-1/2 inch discharge connection located on the pump operator's control panel, two 1-1/2 inch discharge connections located in the 1-1/2 inch transverse hose bed, and one discharge connection to the booster reel. All discharge connections shall have machined ends with a mirror finish to permit hand tight connections. The connection shall provide a seat for discharge hose gaskets, and shall be threaded with NH threads. The 2-1/2 inch connection shall be provided with a rocker lug, chromium plated, gasketed cap chained to the valve body. Each discharge connection shall be provided with a hose discharge pressure gage located on the fire pump operator's control panel. The discharge valves shall be the pump manufacturer's standard valve; and shall be balanced, corrosion resistant, ball type for quarter-turn operation. The 2-1/2 inch valve shall be operated and locked by means of an individual control handle as a single-handed operation. The discharge connections shall be provided with drain valves of at least 3/4-inch id for draining hose lines with the pump discharge gates closed. The push-pull or rotary turn type drain valves shall be of the water service type and shall be equipped with positive stops. Drain-valve controls shall be located adjacent to the discharge connections. Drain lines shall be concealed behind the pump compartment panels and shall discharge to the ground at a point not above the bottom of the chassis frame. Each discharge connection, pressure gage, and drain shall be identified by a related numbering system as well as function.

3.9.4 Relief valve. An automatic relief valve shall be provided for controlling the pressure from all outlets. The location of the relief valve control shall be such that adjustment can readily be made by the operator when standing in normal position, without the use of tools, and in full view of the discharge pressure gage. The relief valve shall be of such design that, when set 10 psi higher than the pump working pressure, there will not be an increase in pump pressure greater than 30 psi when discharge lines are shut slowly. The relief valve shall perform satisfactorily when pumping rated capacity at any pump working pressure over 75 psi. Relief valve drain shall be provided and shall be in a readily accessible location. The relief valve drain line shall discharge to the ground at a point not above the bottom of the chassis frame.

3.9.5 Pressure rating. The pump shall withstand not less than 350 psi hydrostatic pressure on the discharge side for 10 minutes.

3.9.6 Fire pump operator's control panel. The fire truck shall be equipped with a fire pump operator's control panel located on the road side of the truck. The pump pressure and suction gages shall be the silicone liquid filled type equal to Span Instrument Co., Model 507. The pump pressure and suction gages shall be accessible to the operator. All gages shall return to zero reading upon equipment shutdown. All gages and instruments shall be high quality products. All valve operating handles and other controls shall be operable by personnel wearing heavy gloves. All extended controls and handles shall be noninterfering, firmly supported, and realineable. The panel shall include at least the following:

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a. Pump pressure gage 30-inch Hg (vacuum) to 400 psi.
b. Pump suction gage 30-inch Hg (vacuum) to 400 psi.
c. Engine electric tachometer - not less than 4,000 rpm range with a provision in the panel to allow checking accuracy with a test revolution counter.
d. Tank fill valve control.
e. All pump discharge valve controls.
f. Tank suction valve control.
g. Primer pump control.
h. Illumination of all instruments and controls with nonglare lighting.
i. Engine coolant temperature gage.
j. Engine oil pressure gage.
k. Heat exchanger valve control.
l. Five hose discharge pressure gages (0 to 400 rpm), equal to Span Instrument Company, Model 307.
m. Water tank level gage, electronic type.
n. All other pump panel mounted drains and controls mentioned herein.
o. Discharge valve and throttle handles shall be of black extreme temperature resisting plastic, brass, or stainless steel. All gage rims and other exposed metal shall be chromium plated or stainless steel.
p. Plugged 1/4-inch standard pipe connection(s) with pipe or tubing for connection of vacuum or pressure test gage(s) for testing pump pressure and suction gages.
q. Instruction plates specified in 3.21.
r. Pump relief valve.

All controls, valves, drains, and gages shall be identified as to their function and shall have a related identification number. All instruments and switches shall be identified as to their function. All identification plates shall be provided with at least 1/2-inch lettering and shall be attached to the control panel by bolts, sheet metal screws or rivets.

3.10 Water tank. A removable water tank, of not less than 500 gal capacity, shall be provided to the rear of the fire pump housing. The water tank shall not extend beyond the main hose bed sides. The water tank shall be fabricated of either aluminum, copper-bearing steel, or fiberglass reinforced plastic. Aluminum tank construction shall be not less than 3/16 inch and steel tank thickness not less than 10 gage. Fiberglass shall have a design thickness as recommended by the tank manufacturer. Longitudinal and transverse surge partition(s) shall be provided in the tank. The water tank shall be of the horizontal type, with flat top and flat bottom, with sump, and mounted on the truck frame. The water tank shall be capable of discharging 90 percent of its capacity with the fire truck on a 30 percent grade, heading upgrade and also heading downgrade. The top shall be removable, gasketed to prevent leakage, and shall support not less than 200 lb psi of surface area. The tank drain shall be located at the low point of the tank to provide complete drainage. The water tank drain shall discharge directly to the ground at a point not above the bottom of the chassis frame. The tank shall have a non-spash combination overflow and vent pipe of not less than 3-inch nominal pipe size, directed to the ground at the rear of the rear axle, and extending below the bottom of the chassis frame.

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3.10.1 Tank filler opening. The tank shall have an accessibly located filler opening not less than 3 inches in diameter and shall be provided with a tight, gasketed, hinged cap having a spring-type hold-down catch. The tank fill opening shall be provided with a removable strainer.

3.10.2 Tank piping. All piping from the tank to the pump and from the pump to the booster hose reel and transverse hose bed shall be of corrosion-resistant metal, or shall be galvanized. The pipeline between the tank and the pump shall be of adequate size to permit flow rates specified in 3.5 and 3.9 and shall incorporate a flexible, reinforced, noncollapsible connection(s) to take up vibration. A quick-opening tank to pump shutoff valve shall be provided with control extended to the operating side of the truck and a drain valve located at the lowest point with control extended to the operating side of the truck. The tank suction pipeline, valves, and fittings shall withstand not less than 150 psi working pressure. The piping between the pump, booster reel, and transverse hose beds shall be not less than 1-1/2 inch id, and shall incorporate 1-1/2-inch quick-opening valves with controls extended to the pump operator's control panel. The discharge piping, valves, and fittings between the pump booster reel and transverse hose bed shall withstand not less than 500 psi working pressure. A complete piping diagram shall be provided. All drain lines shall discharge to the ground at a point not above the bottom of the chassis frame. The piping system shall provide the capability of removing 90 percent of the water from the water tank when operating upgrade or down-grade on a 30 percent grade.

3.10.3 Tank mounting. The tank shall be supported on sills to cushion shock and transfer the load to the truck frame. The tank shall be so mounted to the chassis that it can be removed without removing the truck body.

3.10.4 Tank level gage. A water tank level gages shall be furnished and located on the fire pump operator's control panel. The gage shall be of the electronic type.

3.11 Hose bed. A hose bed shall be provided with the capability of stowing not less than 500 feet of 2-1/2 inch and 400 feet of 1-1/2-inch double-jacketed, cotton, rubber-lined fire hose. The hose bed interior shall be free of all projections, sharp, or rough edges which might chafe or tear fire hose or hinder its placement or removal. The hose bed sides and end shall be flanged and reinforced to withstand not less than 200 lb horizontal pull at any point on top edge without permanent deformation. The hose bed floor shall be of slatted, well-seasoned, select grade oak, or other hardwood having equal characteristics. The slats shall run the full length of the hose bed, and shall be spaced to provide ventilation for the hose. The hose bed floor assembly shall be removable. Exposed bolt or rivet heads shall be of the oval type. Two folding, spring loaded, chromium plated toe steps shall be furnished, one shall be located on the right, and one on the left, at the rear of water tank or body to permit easy access to the hose bed.

3.11.1 Transverse hose beds. Three transverse hose beds shall be located on the top of the pump compartment area. Two of the transverse beds shall have the capability of stowing in each bed not less than 200 feet of 1-1/2-inch double-jacketed, cotton, rubber-lined fire hose, and the third

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transverse hose bed shall have the capability of stowing 500 feet of 1-inch cotton polyester, rubber-lined forestry hose. Hose payout or removal openings shall be provided on each side of the truck with hose rollers installed in the bottom and both sides of each transverse bed opening to prevent hose interference or snagging. All piping connections to the transverse hose beds shall be 1-1/2 inch with a 1-inch reducer connection to be provided for the hose bed to contain the 1-inch forestry hose. The bottom of the transverse hose beds shall be of slatted, well-seasoned oak, or other hardwood having equal characteristics. Discharge gate controls for each hose bed shall be located at pump control panel, and shall be suitably identified. The hose beds shall be identified as transverse hose bed No. 1, No. 2, and No. 3. Hose bed No. 1 shall be located toward the front of the truck and shall contain 1-1/2-inch hose. Hose bed No. 3 shall contain 1-inch hose. Each hose bed shall be provided with a center mount swivel type discharge connection to permit hose to be payed out from either side of the truck. It is required that the discharge connection be provided space to permit hose attachment to, or removal from, the connection without interference.

3.12 Equipment compartments. All equipment compartments and compartment doors shall be the alloyed coated galvanealed type steel, or aluminum construction and shall be watertight. The cabinets shall be the maximum size practical. Compartment shall be of welded or bolted construction for added strength. Compartment doors shall be double panel construction and shall be attached with full length, heavy-duty stainless steel piano hinges; all vertically hinged doors shall have door hold-open device, equal to the stainless steel, spring type device identified in the industry as moonlite spring. Each compartment door shall be provided with recessed handles operable with heavy-gloved hands with weather-resistant latches. Maximum corrosion-resistant materials shall be utilized in compartment fabrication and hardware. Doors or door opening shall be fully gasketed and weather-stripped with neoprene and all compartment seams shall be complete welds or sealed against water. Drip rails or flanged cover plates shall be provided to deflect rain water from entering compartments. Compartment tops and wheel-housing tops shall run continuously on each side of the truck and shall be of antiskid surface, same as furnished for running boards and rear platform surfaces. All equipment cabinets specified herein shall be 12-gage thick steel, with cabinet doors 14-gage thick steel; or 1/8-inch aluminum with 3/16-inch aluminum doors.

3.12.1 Side equipment compartments. The following side compartments shall be furnished:

a. Street side compartments:

1. One compartment over the rear wheels with a horizontally hinged, fold-down door.
2. One compartment behind and one ahead of the rear wheels with vertically hinged, forward opening door.

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b. Curb side compartments:

1. One compartment over the rear wheels with a horizontally hinged, fold-down door.
2. One compartment behind and one compartment ahead of the rear wheels with vertically hinged, forward opening door.
3. One compartment opposite the street side pump control panel, with vertically hinged, forward opening door, for ready access to pump, piping, and rear of control panel.

3.12.2 Hose reel compartment. A rear hose reel compartment shall be furnished. The hose reel, specified in 3.16, shall be mounted in the center of this compartment. On each side of the hose reel, steel bulkheads shall be provided. The location of the bulkheads shall be such as to provide sufficient space to facilitate servicing and/or removal of the hose reel without bulkhead interference. A vertical roller shall be provided on each side of compartment to protect hose payout from the reel.

3.13 Rear platform and running boards. The fire truck shall be provided with a rear platform and running boards. The rear platform shall have a load-carrying capacity of not less than 500 lb. The rear platform length shall be equal to the width of the truck. The platform shall be not less than 18 inches in width, from front to rear. Rear platform shall have rounded or mitered corners, and shall be flanged downward at least 2 inches. The side running boards shall extend from the rear of the cab to the side equipment cabinets. Running boards shall have a load-carrying capacity of not less than 350 lbs. The rear platform and running boards shall have a steel or aluminum diamond plate surface. The running boards and rear platform shall be located at the same height from ground level. The rear platform and running boards shall be 22 inches +2 inches -0 inches, above ground level, with the truck loaded as specified in 3.6.2.

3.14 Handrails and grabrails. The rear platform shall be provided with two vertical grabrails to assist personnel in climbing onto the rear platform. A horizontal handrail shall be provided at the rear end of the hose bed and above the rear platform. The handrails and grabrails shall be 1-1/4 inch minimum diameter chromium-plated brass, chromium-plated steel, or polished stainless steel tubing. All handrails and grabrails shall be of the nontwist type.

3.14.1 Folding steps. Folding steps shall be provided to assist personnel in gaining access to various firefighting body components. As a minimum, one step shall be provided at the pump operator's panel; one step shall be provided on the curb side directly opposite the pump operator's panel; and two steps shall be provided at the rear, one on each side of the rear platform.

3.15 Buzzer. A buzzer shall be installed in the cab and a weatherproof pushbutton on one of the rear handrail supports to provide for a signaling device from the rear platform to the truck cab.

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3.16 Booster hose reel. A booster hose reel shall be provided and installed in the compartment specified in 3.12.2. The reel shall be an electrical rewind type. The rewind control switch shall be a push type, spring-loaded button, and shall be installed and located near the hose reel. The switch shall be suitably identified. The hose reel shall be equipped with 200 feet, in two 100-foot lengths, and 1 inch, 800 psi, rubber chemical hose, jointed with smooth full-flow type couplings. The handline nozzle shall be of the constant flow variable stream type and shall be fabricated of light-weight "PYROLITE" material or equal. The nozzle shall have a water discharge capacity of 60 gpm at 100 psi. The nozzle shall dispense the water in an infinite variation of patterns between a solid straight stream and full spray. The straight stream shall have an effective reach of at least 60 ft. The handline nozzle shall incorporate a ball or plug type shutoff valve with a horseshoe bail type handle. The female 1-inch fitting shall be of the full-time swivel base. The nozzle shall have a rubber thermoplastic bumper with a double row of molded rubber teeth as an integral part for greater water breakup and finer fog. The hose reel shall have a friction brake payout to prevent overtravel. The hose reel shall also have provisions for manual handcrank rewind, with the handcrank facing toward the fireman standing at the rear of the truck. A handcrank shall be provided. Booster hose reel horizontal and vertical rollers shall be furnished to permit operation on either side of the truck. Rollers shall be spaced to preclude hose couplings from snagging on rollers. The hose reel shall be connected to the discharge manifold of the fire pump with a 1-1/2-inch id galvanized steel pipeline having a flexible section to take up vibration, and shall have an operation valve with handle extended to the left side of the truck on or above the pump operator's control panel. The discharge pipeline, valve, and fittings between the pump and booster reel shall withstand not less than 500 psi working pressure. Drain line shall discharge to the ground at a point not above the bottom of the chassis frame.

3.17 Fire truck equipment. The following equipment shall be provided:

a. Three 8-foot lengths of 3-inch hard suction hose with 3 inch NH long-handled couplers on female ends and 3 inch NH rocker lug couplers on the male ends. The couplers shall be Akron Pyrolite or equal. The suction hose shall conform to the requirements of NFPA 1901. One 3-inch basket type strainer shall be provided.

b. Three hose troughs to accommodate the hoses specified in 3.18a shall be provided on the road side of the body designed for hoses to be stowed with male threaded ends to the rear. The troughs shall be fabricated of the same gage steel as that used for the body. The cross sectional design shall permit ventilation between the hose and the trough. Quick release fastenings shall be provided adjacent to both ends of the hose and shall be of the cam-locking type, so designed, to permit simultaneous release of one end of the three hoses through a single operation of the locking lever. All exposed metal parts of the fasteners shall be chromium plated. The fasteners shall not permit movement of the hose during travel of the truck.

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c. Four 5-gallon "Indian pumps," back packing tanks shall be provided. They shall be mounted on the curbside of the truck above the equipment compartment and wheelhousing. The tanks shall be securely mounted with quick-release fastenings.

3.18 Cleaning, treatment, and painting. The truck's exterior and interior surface shall be cleaned, treated, and painted in accordance with MIL-STD-1223, except as specified herein. The exterior surfaces of the truck cab, chassis and body, and the interior compartment surfaces shall have a finish color of lime yellow, chip No. 13670 of FED-STD-595. The interior surface of the cab shall be painted black with the manufacturer's standard paint. The disc wheels shall be painted the same as the exterior color. Radiator grille, headlight rings, and other exterior trim shall match the exterior color specified except that chrome-plated, cadmium-plated, stainless steel or anodized aluminum surface shall not be painted.

3.19 Identification marking and data plates. Identification marking, registration numbers, and data plates for the truck shall be in accordance with MIL-STD-1223, as specified for the Navy.

3.20 Identification plate. The contracting officer will furnish to the Government inspector the required identification plates. The contractor will be required to stamp the necessary data in the blank spaces thereon and securely affix said plates in a conspicuous place on each unit, assembly or subassembly, and parts as directed by the Government inspector. Nonferrous screws, rivets, or bolts of not less than 1/8 inch in diameter shall be used to affix the plates. Nomenclature shall be "TRUCK, FIREFIGHTING, 250 GAL/MIN, 4X4."

3.21 Instruction plates. The fire truck shall be equipped with instruction plates suitably located, describing any special or important procedures to be followed in operating and servicing the equipment. Plates shall be of a material which will last and remain legible for the life of the equipment, and shall be securely affixed thereto with nonferrous screws or bolts of not less than 1/8-inch diameter. The following plates shall be provided:

a. A plate indicating the fire pump delivery rate in gpm at 150, and 250 psi pressure, with manufacturer's specified engine speeds shall be affixed to the fire pump operator's control panel.
b. All pumping controls, valves, instruments, and such fire truck operating controls whose function is not obvious shall be provided with suitable instruction plates designating function and operation.

3.22 Rustproofing. The truck cab, chassis, and body underside shall be rustproofed in accordance with FED-STD-297.

3.23 Lubrication. Means for lubrication shall be in accordance with the manufacturer's standard practice. The lubricating points shall be easily visible and accessible. Hydraulic lubrication fittings shall be in accordance with SAE J534. Where use of high pressure lubricating equipment, 1,000 psi or higher, will damage grease seals or other parts, a suitable warning shall be affixed to the equipment in a conspicuous location.

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3.24 Servicing and adjusting. Prior to acceptance of the truck by the Government inspector, contractor shall service and adjust each truck and its mounted equipment for operational use, including at least the following: adjustment of lights; adjustment of engine; adjustment of electrical and brake system; alinement of front wheels, inflation of all tires; complete lubrication of chassis, engine, running gear, and mounted equipment.

3.25 Workmanship.

3.25.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.25.2 Bolted connections. Boltholes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts nuts, and screws shall be tight.

3.25.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.25.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.25.5 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the castings ability to perform its intended function.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on one fire truck when a first article is required (see 3.2 and 6.2). This inspection shall include the examination of 4.3 and the tests of 4.4. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3, the tests of 4.4.9, and the packaging inspection of 4.5. This inspection shall be performed on the samples selected in accordance with 4.3.

4.3 Examination. Each truck shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4 Tests.

4.4.1 Truck weight. The truck shall be weighed to determine curb weight and distribution of curb weight on front and rear axle. The imposed loading on front and rear axle will be computed using the curb weight, the operator weight at 175 lb, and the payload uniformly distributed over the vehicle load

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area, to provide the specified gvw. Calculated imposed loads on front and rear axle will be utilized to ascertain that the suspension, and tires furnished are of adequate capacity to meet specification requirements.

4.4.2 Service brakes tests. Service brakes shall be tested to determine conformance to 3.5b and c.

4.4.3 Acceleration test. The truck shall be tested for acceleration to determine conformance to 3.5e.

4.4.4 Road tests. The truck shall be subjected to road tests to determine conformance to 3.5h. Cross-country terrain road test shall utilize 4-wheel drive.

4.4.5 Fire pump capacity and component tests.

4.4.5.1 Pressure test. The pump shall be tested hydrostatically at a pressure of 350 psi for 10 minutes to determine conformance to 3.9.5. The pump pressure test may be performed by the pump manufacturer prior to installation on the truck.

4.4.5.2 Run-in test. A 2 hour run-in test shall be performed in accordance with NFPA 1901. The test shall consist of drafting water with a minimum lift of 10 feet through 20 feet of 3-inch suction hose, and shall include at least the following:

a. One hour at 250 gpm, at 150 psi.
b. One hour at 100 gpm at 250 psi.

4.4.5.3 Capacity tests. Tests shall be conducted in accordance with the test procedures of NFPA No. 1901. All tests shall be performed with calibrated instruments. The following tests shall be performed:

a. The pump shall be capacity tested to determine conformance to 3.9. The pump, while drafting water with a minimum lift of 10 feet through 20 feet of 3-inch suction hose, shall deliver a minimum of 250 gpm at 150 psi net pump pressure for 2 hours, a minimum of 100 gpm at 250 psi net pump pressure for 1/2 hour, and shall be a continuous operational test using one discharge outlet.
b. The pump, while drafting water with a minimum lift of 10 feet through 20 feet of 3-inch suction hose, shall deliver 250 gpm at 165 psi net pump pressure for 10 minutes, and 100 at 275 psi net pump pressure for 10 minutes to verify 10 percent overload.
c. The primer pump shall be tested to determine conformance to 3.9.1. A vacuum test, using the priming pump, with a capped suction at least 24 feet long, shall develop 22 inches of vacuum and hold the vacuum with a drop of not more than 10 inches in 10 minutes. The primer shall not be used after the 1-minute test period has started. The discharge outlets of the pump shall be uncapped.
d. A test shall be conducted to verify that the relief valve performs in accordance with 3.9.4.

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4.4.6 Water tank flow test. The pump shall deliver a minimum of 250 gpm at 150 psi from the water tank for not less than 80 percent of the rated capacity, to determine conformance to 3.9.

4.4.7 Parking brake test. Parking brake test shall be conducted to determine conformance to 3.5a.

4.4.8 Pump and roll test. A pump and roll test shall be performed to determine conformance to 3.5i.

4.4.9 Production truck tests. Each production truck shall be subjected to the tests specified in 4.4.5.2 run-in test; 4.4.5.3b overload test; 4.4.5.3c primer pump test; 4.4.5.3d relief valve test; and 4.4.8 pump and roll test. The results of the tests shall be reported on a form similar to that specified in 4.4.5.3. Each production truck shall be road tested, loaded as specified in 3.6.2, consisting of not less than 5 miles over improved roads a maximum speed of not less than 55 mph; for not less than 5 miles over unimproved roads at a maximum speed of not less than 30 mph; and not less than 5 miles over cross-country terrain at a maximum speed of not less than 15 mph utilizing 4-wheel drive. Failure of any test shall be cause for rejection.

4.5 Packaging inspection. The vehicle shall be inspected to verify conformance to the requirements of section 5.

5. PACKAGING

5.1 Vehicle processing. The equipment shall be preserved and packed in accordance with the contractor's standard practice. When specified (see 6.2), equipment shall be preserved and packed in accordance with the mobile requirements of MIL-STD-281 with the level of preservation and packing required.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

* 6.1 Intended Use. Four wheel drive fire trucks are intended to fight brush and grass fires where accessibility by larger fire fighting equipment is not possible.

* 6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.1.2).
c. When first article is required for inspection and approval (see 3.2, 4.2.1).
d. When quality assurance inspection and tests are required (see 3.2.1).

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e. When gasoline is required (see 3.7.1.2).
f. When preservation and packing shall be in accordance with MIL-STD-281 with level of preservation and packing required (see 5.1).

6.3 Data requirements. When this document is used in an acquisition and data are required to be delivered, the data requirements shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD FAR Supplement, Part 27, Sub-Part 27.475-1 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data should be delivered by the contractor in accordance with the contract or purchase order requirements.

6.4 First article. When a first article inspection is required, the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one fire truck. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

* 6.5 Subject term (key word) listing.

Diesel engine
Fire fighting vehicle
Fire pump
Water tank

6.6 Changes from previous issue. The margins of this document are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Navy - YD
Army - ME

User Activity:
Navy - MC

Preparing Activity:
Navy - YD

(Project 4210-0392)

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INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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DEPARTMENT OF THE NAVY

Commanding Officer (Code 156)
Naval Construction Battalion Center
Port Hueneme, CA 93043-5000

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-29185B	Truck, Firefighting: Brush/Grass, 18,000 Pounds GVW, 4 By 4, Commercial Chassis

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 1 82 MAR

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